

1 Causal inference in latent space

We construct a Bayesian causal-inference model for multivariate time series. The goal is to identify a causal relationship between time series given the constructed model. We suppose that the unknown dynamics generated the observable time series from the latent space.

There given source and target time series $\mathbf{x}_t$ and $\mathbf{y}_t$ as observations. The time series are generated by a discrete-time dynamical system in the latent space $\mathbf{z}_t$ with unknown dynamics:

$$\mathbf{y}_{t+1} = f(\mathbf{z}_t, \mathbf{x}_t, \mathbf{y}_t) + \boldsymbol{\zeta}_{t+1},$$

The time series are represented via *delay embedding*. The matrix of delay vectors for the scalar-valued series x_t is the Hankel matrix:

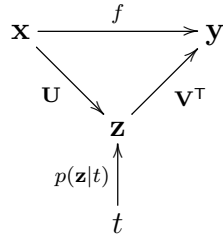
$$\mathbf{X} = \begin{pmatrix} x_{t-1} & x_{t-2} & \cdots & x_{t-(N+T)} \\ x_{t-2} & x_{t-3} & \cdots & x_{t-(N+T-1)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{t-N} & x_{t-(N-1)} & \cdots & x_T \end{pmatrix},$$

where $T \ni \{1, \dots, t\}$ is the sample size, N is the original embedding dimension, and the time lag $\Delta t = 1$. A column of this matrix contains a point $\mathbf{x}_t$ in the state space. The sequence $\{\mathbf{x}_t\}$ forms the phase trajectory $\mathbf{X}$. Call the *embedding* the linear map $\mathbf{z} = U\mathbf{x}$.

Set the forecasting model

$$f: \mathbf{x} \rightarrow \mathbf{y}$$

as a linear operator. Introduce the singular value decomposition $f = U\Lambda V^T$:



The system dynamics is represented by the generative model $p(\mathbf{z})$ that approximates the embedding in the latent space with the support t .

Treat this superposition as a decoder and a encoder, where the linear operators U and V are the orthogonal and Λ is diagonal with positive elements. Each row of the matrix

$$U\mathbf{x} = \begin{pmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \\ \vdots \\ \mathbf{u}_k \end{pmatrix}^T \mathbf{x}$$

produces an element of a vector $\mathbf{z}$ in the latent space by the dot product

$$z_i = \mathbf{u}_i^T \mathbf{x}.$$

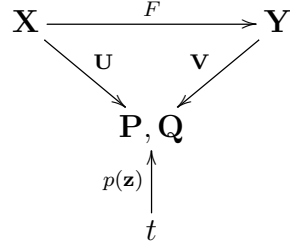
1.1 The probabilistic view

Let the prior distribution of a vector $\mathbf{u} \sim \mathcal{N}(\mathbf{u}, \mathbf{A})$. Its parameters are estimated as empirical distribution from the data.

2 Computational experiment

For control problem we investigate a driven pendulum problem. A pendulum of a given length has its own period of oscillation. Driven by an external force a pendulum had forced oscillation. These two oscillations are simple and independent. They superposition are observed. A change in the frequency of the driving force disturbs the superposition. The goal is to identify the causal relationship between the driving force and the normal mode of the pendulum oscillations.

3 Appendix: Control problem and Canonical Correlation Analysis



4 Appendix: countinous time dynamical system and delay embedding

For the continuous-time dynamical system, the state vector $\mathbf{z}(t)$ evolves according to the differential equation

$$\frac{d\mathbf{z}(t)}{dt} = f(\mathbf{z}(t), \mathbf{x}(t), \mathbf{y}(t)) + \boldsymbol{\varepsilon}(t),$$

and the observed time series are generated from the state vector via the observation functions:

$$\mathbf{x}(t) = h_x(\mathbf{z}(t)) + \boldsymbol{\xi}_x(t), \quad \mathbf{y}(t) = h_y(\mathbf{z}(t)) + \boldsymbol{\xi}_y(t),$$

where $\boldsymbol{\varepsilon}(t)$, $\boldsymbol{\xi}_x(t)$, and $\boldsymbol{\xi}_y(t)$ are noise terms.

The matrix of delay vectors for the scalar-valued series x_t is the Hankel matrix:

$$\mathbf{X} = \begin{pmatrix} x(t_1) & x(t_2) & \cdots & x(t_{N-T}) \\ x(t_1 - \Delta t) & x(t_2 - \Delta t) & \cdots & x(t_{N-T} - \Delta t) \\ \vdots & \vdots & \ddots & \vdots \\ x(t_1 - (T-1)\Delta t) & x(t_2 - (T-1)\Delta t) & \cdots & x(t_{N-T} - (T-1)\Delta t) \end{pmatrix},$$

where T is the length of the time series sample, $t \in (1, T)$. The time lag is Δt and N is the embedding dimension.

The continuous-time forecasting model is a convolution of two functions: the input signal $x(t)$ and the kernel $K(t)$ that represents the system dynamics. Thus, the output signal $y(t)$ can be expressed in the kernel terms:

$$y(t) = \int_{t-T}^t K(t - \tau)x(\tau)d\tau + \varepsilon(t).$$

The vector form of this equation is:

$$\mathbf{y}_t - \boldsymbol{\varepsilon}_t = \mathbf{K}^\top \mathbf{x}_t,$$